

Pack expansions in *using-declarations*

ISO/IEC JTC1 SC22 WG21, Core Working Group

P0195R2

using-declarations should be able to cope with lists and pack expansions to allow flexible injection of names.

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2016-11-08

Motivation

With variadic templates having been introduced in C++11, many language constructs were updated to operate on pack expansions. Just as some situations require derivation from a pack of classes, some require introduction of a name from a pack of classes. Consider

```
template <typename... T>
struct Overloader : /* [...] */ {
    // [...]
};

template <typename... T>
constexpr auto make_overloader(T&&... t) {
    return Overloader<T...>{std::forward<T>(t)...};
}

int main() {
    auto o = make_overloader([] (auto const& a) {std::cout << a;},
                             [] (float f) {std::cout << std::setprecision(3) << f;});
}
```

The implementation of `Overloader` is verbose and inefficient; One has to recursively introduce overloads of `operator()`:

```
template <typename T, typename... Ts>
struct Overloader : T, Overloader<Ts...> {
    using T::operator();
    using Overloader<Ts...>::operator();
    // [...]
};

template <typename T> struct Overloader<T> : T {
    using T::operator();
};
```

While it's possible to achieve logarithmic instantiation depth, the resulting implementation is needlessly complicated and still less efficient than the following, terse syntax:

```
template <typename... Ts>
struct Overloader : Ts... {
    using Ts::operator()...;
    // [...]
};
```

Design decisions and impact

We propose that a *using-declaration* consist of a list of names, each called a *using-declarator*. As the proposed change is solely an enhancement, well-formed programs and their semantics are unaffected.

Proposed wording

Modify 3.3.2 [basic.scope.pdecl] ¶4 as indicated:

The point of declaration of a *using-declaration* *declarator* that does not name a constructor is immediately after the *using-declaration* *declarator* (7.3.3).

Modify 3.4.3.1 [class.qual] ¶2 (2.2) as indicated:

— in a *using-declarator* of a *using-declaration* (7.3.3) that is a *member-declaration*, if the name specified after the *nested-name-specifier* is the same as the *identifier* or the *simple-template-id*'s *template-name* in the last component of the *nested-name-specifier*,

Modify 7.3.3 [namespace.udecl] ¶1 as indicated:

A *using-declaration* introduces a set of declarations into the declarative region in which the *using-declaration* appears.

using-declaration:

using *using-declarator-list* ;

using-declarator-list:

using-declarator ...*opt*

using-declarator-list , *using-declarator* ...*opt*

using-declarator:

typename_{opt} *nested-name-specifier* *unqualified-id* ⁺

Each *using-declarator* in a *using-declaration*⁹⁸ introduces a set of declarations into the declarative region in which the *using-declaration* appears. The set of declarations introduced by the *using-declarationdeclarator* is found by performing qualified name lookup (3.4.3, 10.2) for the name in the *using-declarationdeclarator*, excluding functions that are hidden as described below. If the *using-declarationdeclarator* does not name a constructor, the *unqualified-id* is declared in the declarative region in which the *using-declaration* appears as a synonym for each declaration introduced by the *using-declarationdeclarator*. [*Note*: Only the specified name is so declared; specifying an enumeration name in a *using-declaration* does not declare its enumerators in the *using-declaration*'s declarative region. — *end note*] If the *using-declarationdeclarator* names a constructor, it declares that the class inherits the set of constructor declarations introduced by the *using-declaration* from the nominated base class.

98) A *using-declaration* with more than one *using-declarator* is equivalent to a corresponding sequence of *using-declarations* with one *using-declarator*.

Modify 7.3.3 [namespace.udecl] ¶3 as indicated:

In a *using-declaration* used as a *member-declaration*, the *using-declarator's* *nested-name-specifier* shall name a base class of the class being defined. If such a *using-declarationdeclarator* names a constructor, the *its* *nested-name-specifier* shall name a direct base class of the class being defined. [*Example*:

```
template <typename... bases>
struct X : bases... {
    using bases::g...;
};

X<B, D> x; // OK: B::g and D::g introduced
```

— *end example*]

Modify 7.3.3 [namespace.udecl] ¶10 as indicated:

[*Example*:

```
namespace A {
    int i;
}

namespace A1 {
    using A::i;
    using A::i, A::i; // OK: double declaration
}

struct B {
    int i;
};

struct X : B {
    using B::i;
    using B::i, B::i; // error: double member declaration
};
```

— *end example*]

Modify 7.3.3 [namespace.udecl] ¶19 as indicated:

If a *using-declarationdeclarator* uses the keyword *typename* and specifies a dependent name (14.6.2), the name introduced by the *using-declaration* is treated as a *typedef-name* (7.1.3).

Add a bullet (4.13) to the list in 14.5.3 [temp.variadic] ¶4:

— In a *using-declaration* (7.3.3); the pattern is a *using-declarator*.

Add a bullet (8.4.2) to the list in 14.6 [temp.res] ¶8:

— lookup for a name in the template definition found a *using-declaration*, but the lookup in the corresponding scope in the instantiation does not find any declarations because the *using-declaration* was a pack expansion and the corresponding pack is empty.

SD-6 Feature Test Macro

The feature test macro `__cpp_variadic_using` is recommended for this feature.

Acknowledgments

Thanks to Daveed Vandevoorde for assistance in preparing the wording.